25900750

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCBTG100 - ECOLOGY ANDENVIRONMENTAL SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer the Questions in One Word or Sentence-Answer All Questions

Each Question carries 1 mark

1. Name the gas that plants absorb during photosynthesis [K] / [CO1]
   Carbon dioxide- 1 mark
2. What habitat restoration technique is commonly used in desert ecosystems? [K] / [CO3]
   Any one- 1 mark
3. Who launched CAMPA in 2009? [K] / [CO4]
   Supreme court- 1 mark
4. What is the main goal of the Nilgiri Tahr Project (2023)? [K] / [CO4]
   Conservation- 1 mark
5. Which abiotic factor determines the clarity of water in lakes and rivers? [U] / [CO1]
   Any one (temperature/pH etc)- 1 mark
6. Define Denitrification. [K] / [CO1]
   Definition- 1 mark
7. What is the importance of ecosystem conservation? [U] / [CO2]
   Any two points-1 mark
8. Write down the types of conservation strategies. [K] / [CO2]
   Any two strategies- 1mark
9. Give an example for an aquatic ecosystem. [K] / [CO1]
   Any one example- 1 mark
10. What shape represents energy flow in ecosystems? [K] / [CO1]
    Pyramid- 1 mark

[10x1 = 10]

Part B

Short Answer Questions-Answer any 3 Questions

Each Question carries 3 marks

11. What is the role of decomposers in ecosystem processes? [K] / [CO1]
    Decomposer definition – 1 mark, Role – 2 marks
12. Describe the objectives of NRCP. [K] / [CO4]
    Expand the term – 1mark, Any two objectives – 2 marks
13. How does reforestation help in climate change mitigation? [U] / [CO4]
    Any three points – 3marks
14. Recall the significance of carbon cycle in an ecosystem. [K] / [CO1]
    Carbon cycle-Definition – 1mark, Significance-Any two points – 2 marks
15. What are biotic and abiotic factors? [K] / [CO1]
    Biotic and abiotic factors - Definition with examples – 3 marks

[3x3 = 9]

Part C

Short Essay Questions-Answer any 1 Question

Each Question carries 6 marks

16. Describe the impact of deforestation on ecosystem services. [U] / [CO1]
    Deforestation-Introduction – 1mark, impact of deforestation on ecosystem services -Any 5 points – 5 marks
17. Discuss the ecological importance of the Sundarbans mangrove ecosystem. [A] / [CO4]
    Sundarbans mangrove ecosystem, Importance- explain (six points)- 6 marks

[1x6 = 6]

Part D

Essay Questions-Answer any 1 Question

Each Question carries 10 marks

18. Discuss characteristics of ecological succession. Briefly explain the causes. [A] / [CO2]
    Ecological succession-Definition - 2 marks, Types / characteristics and mechanisms – 5 marks
    Causes – 3 marks
19. Analyze the long-term ecological impacts of the Arabian Oryx reintroduction on desert ecosystems.[An] / [CO3]
    Arabian Oryx – introduction/significance-2 marks, Global Restoration project -introduction, Description – 3 marks, Details of reintroduction – 5 marks

[1x10 = 10]